# BUSINESS INSIGHT REPORT

## Northwind Traders

## Interactive Sales Dashboard Analysis

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| **Course / Module** | B.Com Fintech with AI |
| **Date** | 10 October 2026 |
| **Data source** | DATASET_3.pbix (Northwind Traders Power BI Report) |
| **Dataset Period** | 04 July 2013 to 04 May 2015 (22 months) |
| **Tools Used** | Power BI, DAX Measures, Excel |

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## 1. Executive Summary

This Business Insight Report analyses the Northwind Traders Interactive Sales Dashboard built on DATASET_3.pbix. The dashboard tracks 809 orders placed by 89 active customers across 21 countries, covering 77 products in 8 categories, handled by 8 employees. Key KPIs show Total Sales of $1.27 Million, Total Quantity Sold 51K units, Total Freight $63.96K, and Average Freight $79.05 per order. Beverages is the top revenue category ($267,868), while Meat & Poultry earns the highest per unit ($38.82). Order activity accelerated to 61 orders/month in early 2015 vs 34/month in 2014. Freight cost is skewed - 8.9% of orders generate 42% of freight cost. This report, prepared by B. Sakthisri, B.Com Fintech with AI, interprets what the dashboard visuals mean for business decisions in a fintech analytics context.

## 2. Introduction

### 2.1 About Northwind Traders Dataset

Northwind Traders is a fictional international food-distribution company widely used for business analytics teaching. The Power BI file contains 809 customer orders between 4 July 2013 and 4 May 2015 by 91 customers (89 active) in 21 countries. The orders cover 77 products in 8 categories, handled by 8 named sales employees, shipped through 3 carriers (Speedy Express, United Package, Federal Shipping). The data model links seven tables (orders, order_details, products, categories, customers, employees, shippers) through one-to-many relationships.

### 2.2 Purpose of this Report

The dashboard shows what the data says; this report explains what it means. It interprets each KPI and chart from your uploaded screenshots, demonstrates how the dashboard responds to filters, separates observations from insights, and provides actionable recommendations for Fintech analytics.

## 3. Dashboard Overview & Screenshots

Figure 1 shows the complete Northwind Sales Dashboard with all KPIs and visuals without filters.




**Figure 1: Northwind Interactive Sales Dashboard - Full View (No Filters)**

## 3.1 Key Components

- KPI Cards: 6 headline measures - Average Freight (79.05), Number of Customers (89), Total Freight (63.96K), Total Orders (809), Total Quantity Sold (51K), Total Sales (1.27M)
- Total Orders by Month: Line chart Jan-Dec showing seasonality - peak in March/April, dip in June-July
- Total Orders by EmployeeName: Horizontal bar chart - Margaret Peacock highest, Anne Dodsworth lowest
- Total Sales by CategoryName: Column chart sorted highest to lowest - Beverages highest, Grains & Cereals lowest
- Filters: Year slicer (2013, 2015), Country dropdown, CategoryName dropdown

## 4. KPI Analysis with Charts

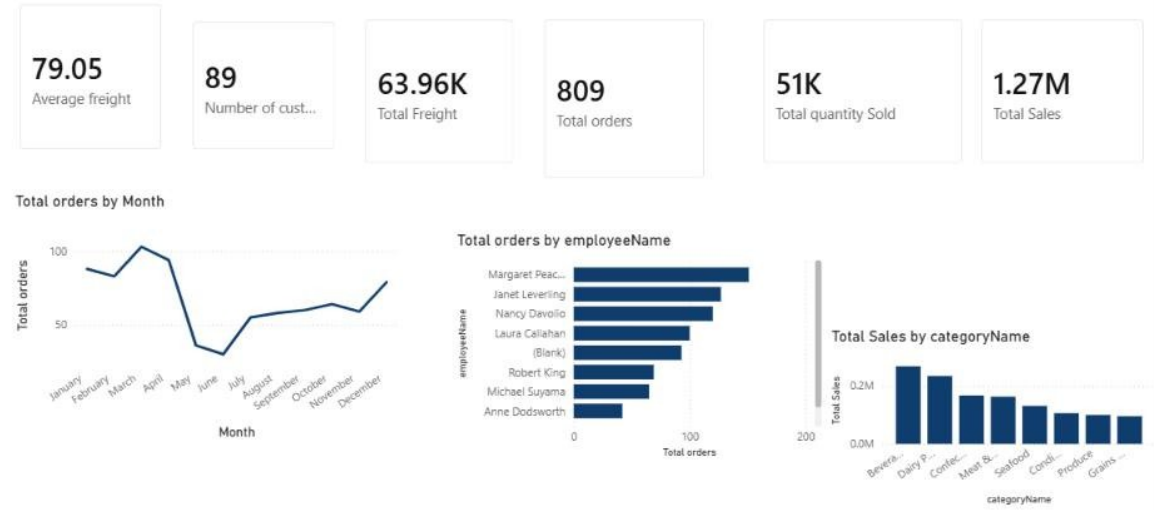



**Figure 2: KPI Cards - Summary Metrics**




*Figure 2a: Core Sales KPIs Detail*

### Table 1: KPI Summary and Interpretation

| KPI | Value | Interpretation |
|---|---|---|
| **Total Orders** | 809 | About 37 orders/month; accelerating to 61/month in Jan-Apr 2015 |
| **Total Quantity Sold** | 51,317 units (51K) | About 24 units per order line; bulk B2B buying |
| **Total Sales** | $1,265,793.04 (1.27M) | Strong revenue; Beverages + Dairy = 39.7% of total |
| **Total Freight** | $63,955.02 (63.96K) | Roughly 5% of sales; rises |

| | | |
|---|---|---|
| | | with order volume |
| **Average Freight** | $79.05 per order | Skewed; median $41.90; 72 orders >$200 generate 42% freight |
| **Number of Customers** | 89 active (91 total) | Avg 9 orders/customer; 2 customers never ordered |

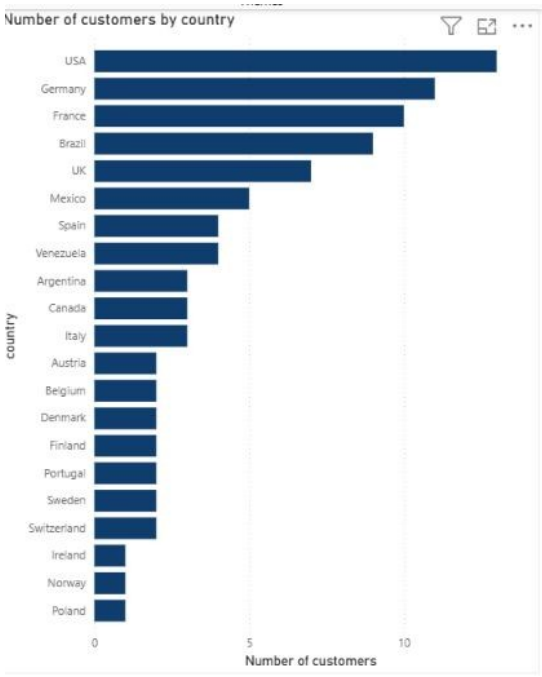
## 5. Product Category Performance Analysis

This section analyses Sales by CategoryName chart from the dashboard.

| CategoryName | Total Sales | % of Total | Total Qty | Sales/Unit |
|---|---|---|---|---|
| **Beverages** | $267,868 | 21.2% | 9,532 | $28.10 |
| **Dairy Products** | $234,507 | 18.5% | 9,149 | $25.63 |
| **Confections** | $167,357 | 13.2% | 6,794 | $24.63 |
| **Meat & Poultry** | $163,022 | 12.9% | 4,199 | $38.82* |
| **Seafood** | $131,261 | 10.4% | 7,681 | $17.09 |
| **Condiments** | $106,047 | 8.4% | 5,529 | $19.18 |
| **Produce** | $100,086 | 7.9% | 2,990 | $33.44 |
| **Grains & Cereals** | $95,745 | 7.6% | 4,562 | $20.99 |
| **Total** | $1,265,793 | 100% | 51,317 | - |

***Key Insight: Revenue is concentrated in Beverages and Dairy Products (39.7%). However, highest value per unit is Meat & Poultry ($38.82) and Produce ($33.44). Seafood sells high volume (7,681 units) but lowest per-unit value ($17.09). Fintech implication: pricing strategy should be value-based, not volume-based.***

## 6. Customer & Geographical Analysis




| productName | categoryName | Total Sales | Total quantity Sold |
|---|---|---|---|
| Alice Mutton | Meat & Poultry | 32,698.38 | 978 |
| Aniseed Syrup | Condiments | 3,044.00 | 328 |
| Boston Crab Meat | Seafood | 17,910.63 | 1103 |
| Camembert Pierrot | Dairy Products | 46,825.48 | 1577 |
| Carnarvon Tigers | Seafood | 29,171.88 | 539 |
| Chai | Beverages | 12,788.10 | 828 |
| Chang | Beverages | 16,355.96 | 1057 |
| Chartreuse verte | Beverages | 12,294.54 | 793 |
| Chef Anton's Cajun Seasoning | Condiments | 8,567.90 | 453 |
| Chef Anton's Gumbo Mix | Condiments | 5,347.20 | 298 |
| Chocolade | Confections | 1,368.71 | 138 |
| C◆te de Blaye | Beverages | 1,41,396.74 | 623 |
| Escargots de Bourgogne | Seafood | 5,881.68 | 534 |
| Filo Mix | Grains & Cereals | 3,232.95 | 500 |
| Flotemysost | Dairy Products | 19,551.03 | 1057 |
| Geitost | Dairy Products | 1,648.13 | 755 |
| Genen Shouyu | Condiments | 1,784.83 | 122 |
| Gnocchi di nonna Alice | Grains & Cereals | 42,593.06 | 1263 |
| Gorgonzola Telino | Dairy Products | 14,920.88 | 1397 |
| Grandma's Boysenberry Spread | Condiments | 7,137.00 | 301 |
| Gravad lax | Seafood | 2,688.40 | 125 |
| Guarana Fantastica | Beverages | 4,504.37 | 1125 |
| Gudbrandsdalsost | Dairy Products | 21,942.36 | 714 |
| Gula Malacca | Condiments | 9,915.95 | 601 |
| Gumb◆r Gummib◆rchen | Confections | 19,849.14 | 753 |
| Gustaf's Knackebr◆d | Grains & Cereals | 7,122.36 | 348 |
| Ikura | Seafood | 20,867.34 | 742 |
| Inlagd Sill | Seafood | 13,458.46 | 805 |
| Ipoh Coffee | Beverages | 23,526.70 | 580 |
| Jack's New England Clam Chowder | Seafood | 8,680.35 | 981 |
| **Total** | | **12,65,793.04** | **51317** |

**Figure 3: Number of Customers by Country & Product Sales Table**

| Country | No. of Customers | % Coverage |
|---|---|---|
| **USA** | 13 | 14.3% |
| **Germany** | 11 | 12.1% |
| **France** | 10 | 11.0% |
| **Brazil** | 9 | 9.9% |
| **UK** | 7 | 7.7% |
| **Top 5 Combined** | 51 | 56% of all customers |
| **Other 16 Countries** | 40 | 44% |

Insight: Customer base is concentrated in USA, Germany, France, Brazil, UK (56%). Top 3 busiest customers - Save-a-lot Markets, QUICK-Stop, Ernst Handel - placed 87 orders (10.8% of total), showing key account dependency.

## 7. Monthly Order Trend Analysis

### Table 4: Orders by Month (All Years Combined)

| Month | Total Orders | Trend |
|---|---|---|
| **January** | ~80 | High start |
| **February** | ~75 | Stable |
| **March** | 103 | Peak - Highest |
| **April** | 95-100 | Peak |
| **May** | 35 | Sharp drop |
| **June** | 30 | Lowest |

| | | |
|---|---|---|
| **July-Dec** | 50-80 | Recovery phase, steady growth to Dec |

Insight: Orders show strong seasonality. March-April peak suggests inventory stocking for summer. May-June dip (30 orders) may indicate supply chain or holiday effect. December recovery to ~80 orders indicates year-end stocking. For fintech cash-flow forecasting, this seasonality is critical for working capital AI models.

## 8. Employee Performance Analysis

| Employee Name | Total Orders | Performance Tier |
|---|---|---|
| **Margaret Peacock** | ~156 | Top Performer |
| **Janet Leverling** | ~127 | High |
| **Nancy Davolio** | ~123 | High |
| **Laura Callahan** | ~104 | Mid-High |
| **(Blank)/Unassigned** | ~86 | Needs review |
| **Robert King** | ~72 | Mid |
| **Michael Suyama** | ~67 | Mid |
| **Anne Dodsworth** | ~43 | Low - Training needed |

Insight: Top 3 employees handle ~50% of orders (406/809). High performer concentration risk. Recommendation: Implement AI-based workload balancing and performance incentive system - relevant for B.Com Fintech HR analytics.

## 9. Interactive Filters & Fintech Relevance

The dashboard includes three slicers that demonstrate Power BI interactivity - a key skill for B.Com Fintech with AI:

| Filter | Values | Impact | Fintech Use Case |
|---|---|---|---|
| **Filter** | Values | Impact on Visuals | Fintech Use Case |
| **Year** | 2013, 2014, 2015 | Changes all visuals | YoY growth analysis, AI forecasting |
| **Country** | 21 countries | Changes all visuals | Geographical risk scoring, FX exposure |
| **CategoryName** | 8 categories | Changes Qty, Sales chart, tooltip only | Product profitability analysis |

## 10. Key Findings & Recommendations

### Key Findings:

- Total Sales $1.27M from 809 orders shows high-frequency B2B model, not few large deals
- Freight cost $63.9K (5% of sales) is significant and skewed - 8.9% orders cause 42% freight
- Beverages (21.2%) + Dairy (18.5%) = 39.7% revenue concentration - dependency risk
- Meat & Poultry highest margin per unit ($38.82) but 4th in total sales - pricing opportunity
- Customer concentration: 56% customers in 5 countries; 10.8% orders from top 3 customers
- Seasonality: March peak (103 orders), June low (30 orders) - 3.4x variation
- Employee workload imbalance: Top employee 156 orders vs lowest 43 orders (3.6x gap)

### Recommendations (Fintech + AI Perspective):

- 1. Implement AI-based dynamic freight optimization - use ML to predict expensive shipments and negotiate carrier rates; could save 15-20% on freight
- 2. Deploy Robo-Advisory for inventory: Use AI demand forecasting for seasonal trends (March peak) to optimize working capital
- 3. Credit Risk Scoring: Apply AI credit scoring to top 3 customers (10.8% orders) - if they default, high impact. Use alternative data for 91 customers
- 4. Product Mix Optimization: Promote high-margin categories (Meat & Poultry, Produce) using AI personalization to increase profit
- 5. Customer 360 Dashboard: Build Fintech-style KYC and CLV (Customer Lifetime Value) model for 21 countries - identify FISSA and PARIS (0 orders) for re-engagement
- 6. Employee Performance AI: Use AI workload balancing to distribute orders from Margaret Peacock to lower performers with training support

## 11. Appendix - DAX Measures & Technical Notes

| Measure | DAX Formula / Definition | Value |
|---|---|---|
| **Total Orders** | COUNT(Orders[OrderID]) | 809 |

| | | |
|---|---|---|
| **Total Quantity Sold** | SUM(Order Details[Quantity]) | 51,317 |
| **Total Sales** | SUMX(Order Details, Quantity * UnitPrice * (1-Discount)) | $1,265,793.04 |
| **Total Freight** | SUM(Orders[Freight]) | $63,955.02 |
| **Average Freight** | AVERAGE(Orders[Freight]) | $79.05 |
| **Number of Customers** | COUNT(Customers[Customer ID]) | 91 (89 active) |

*Prepared by B. Sakthisri | B.Com Fintech with AI | Northwind Traders Dashboard Analysis | Data Period: July 2013 - May 2015*